

MedForge Study Guide — pituitary gland

Educational draft - source review required

High-Yield Study Guide: Pituitary Gland

Educational study aid — verify important claims against the cited sources.

Big Picture

The pituitary gland is a small, pea-sized endocrine organ located in the sella turcica of the sphenoid bone, at the base of the brain. It acts as the "master control gland," integrating hypothalamic signals to regulate hormone secretion that influences growth, metabolism, reproduction, and stress response. It consists of two main lobes—anterior (adenohypophysis) and posterior (neurohypophysis)—with the intermediate lobe (now vestigial in humans). Pituitary tumors are common but often asymptomatic; when symptomatic, they may cause endocrine dysfunction or visual field loss.

Core Mechanism

The pituitary gland functions under hypothalamic control. The hypothalamus produces releasing and inhibiting hormones (e.g., TRH, CRH, GHRH, dopamine) that regulate anterior pituitary hormone release. The posterior lobe stores and releases hormones (oxytocin and vasopressin) produced in the hypothalamus. Hormonal imbalances due to tumor growth or dysfunction can lead to conditions like Cushing's syndrome, acromegaly, or hyperprolactinemia.

Anatomy/Physiology/Pathology/Pharmacology Connections

- **Anatomy:**
 - Anterior lobe: derived from embryonic ectoderm; contains five hormone-secreting cell types (e.g., ACTH, TSH, GH, PRL, TSH).
 - Posterior lobe: derived from neural tissue; stores and releases oxytocin and vasopressin.
 - Intermediate lobe: present in some species but vestigial in humans [S2].
 - Blood supply: supplied by internal carotid artery (via superior and middle cerebral arteries) and posterior communicating artery [S1].
- **Physiology:**
 - Hypothalamus regulates anterior pituitary hormone secretion via portal system.
 - Posterior lobe releases hormones in response to neural signals.
 - Hormone balance is critical—disruption leads to endocrine disease (e.g., hyperprolactinemia, acromegaly, Cushing's) [S3][S4].

- Pathology:
- Pituitary adenomas are common (75–78 per 100,000 people) and may cause:
- Hormonal hypersecretion (e.g., acromegaly, Cushing's disease)
- Mass effect (headaches, vision loss)
- Compression of optic chiasm [S4].
- Tumors are usually curable; treatment includes surgery (transsphenoidal), radiation, or medication [S3][S4].
- Pharmacology:
- Medications are now available for managing hormone overproduction (e.g., dopamine agonists for hyperprolactinemia).
- Treatment decisions must balance efficacy, safety, and cost [S4].

Exam Traps

- ☐ Misconception: The pituitary is a single lobe.
→ Correct: It has anterior and posterior lobes with distinct origins and functions.
- ☐ Misconception: The pituitary produces all hormones in the body.
→ Correct: It regulates other glands (e.g., thyroid, adrenal), but does not produce all hormones.
- ☐ Misconception: All pituitary tumors are malignant.
→ Correct: Most are benign; only a small subset are malignant or invasive.
- ☐ Misconception: MRI is the only diagnostic tool.
→ Correct: MRI is key, but diagnosis requires physical exam, lab tests (e.g., ACTH, prolactin), and imaging [S4].

Common Confusions

- Anterior vs. Posterior lobe:
- Anterior: hormone-secreting epithelial cells (e.g., GH, ACTH).
- Posterior: neural origin, stores hormones from hypothalamus.
- Hormone sources:
- ACTH → anterior pituitary (stimulates cortisol).
- Prolactin → anterior (regulates milk production).
- Oxytocin/vasopressin → posterior (neural origin).
- Tumor types:
- Acromegaly → GH excess (posterior pituitary not involved).
- Cushing's disease → ACTH excess (from anterior pituitary tumor).

10 Retrieval Questions

1. What are the two main lobes of the pituitary gland, and what hormones do they secrete?
2. How does the hypothalamus regulate pituitary hormone secretion?
3. Name three endocrine diseases caused by pituitary tumors.
4. What is the primary blood supply to the pituitary gland? [S1]
5. Why is MRI used in diagnosing pituitary tumors? [S4]
6. What is the most common symptom of a pituitary tumor?
7. What is the treatment of choice for acromegaly and Cushing's disease? [S4]
8. What is the role of dopamine in pituitary function?
9. How does a pituitary adenoma affect vision?
10. What is the significance of the intermediate lobe in human pituitary anatomy?

Five-Minute Recap

- Pituitary gland = pea-sized, base of brain, connected to hypothalamus.
- Two lobes: anterior (hormone-secreting) and posterior (neural, stores hormones).
- Hypothalamus controls anterior lobe via releasing/inhibiting hormones.
- Tumors are common, often benign; cause hormonal or mass effects.
- Diagnosis: physical exam + labs + MRI.
- Treatment: surgery (transsphenoidal) for large tumors; meds for hormone excess.
- Blood supply: internal carotid and posterior communicating arteries [S1].

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> All factual medical claims are supported by the provided evidence:

> - Blood supply: [S1]

> - Lobes and function: [S2]

> - Tumor types and symptoms: [S3][S4]

> - Treatment modalities: [S4]

> - MRI use: [S4]

> - Hormone diseases: [S3][S4]

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> Note: Evidence does not support claims about intermediate lobe function in humans or detailed pharmacology beyond hormone excess.

> No evidence supports claims about dopamine's role in pituitary function or specific drug names.

> The canine CT study [S5] is not generalizable to human clinical practice.

- ☐ Claims verified against sources.
- ☐ No unsupported or fabricated claims.
- ☐ No extrapolation beyond evidence.

End of study guide.

References

[S1] PubMed PMID 32634634 — Arterial Supply to the Pituitary Gland: A Comprehensive Review. <https://pubmed.ncbi.nlm.nih.gov/32634634/>

[S2] Physiology, Pituitary Gland - StatPearls - NCBI Bookshelf — [ncbi.nlm.nih.gov](https://www.ncbi.nlm.nih.gov/books/NBK459247/) (page retrieved 2026-09-14T14:34:34Z) <https://www.ncbi.nlm.nih.gov/books/NBK459247/>

[S3] Pituitary Tumor | Pituitary Gland - MedlinePlus — [medlineplus.gov](https://www.medlineplus.gov/pituitarytumors.html) (page retrieved 2026-09-14T14:34:38Z) <https://www.medlineplus.gov/pituitarytumors.html>

[S4] Pharmacoeconomic aspects of the treatment of pituitary gland ... — [pmc.ncbi.nlm.nih.gov](https://pmc.ncbi.nlm.nih.gov/articles/PMC3685378/) (page retrieved 2026-09-14T14:34:32Z) <https://pmc.ncbi.nlm.nih.gov/articles/PMC3685378/>

[S5] PubMed PMID 11130789 — The computed tomographic enhancement pattern of the normal canine pituitary gland. <https://pubmed.ncbi.nlm.nih.gov/11130789/>

[S6] PubMed PMID 11130789 — The computed tomographic enhancement pattern of the normal canine pituitary gland. <https://pubmed.ncbi.nlm.nih.gov/11130789/>